Lecture 6: Review Questions

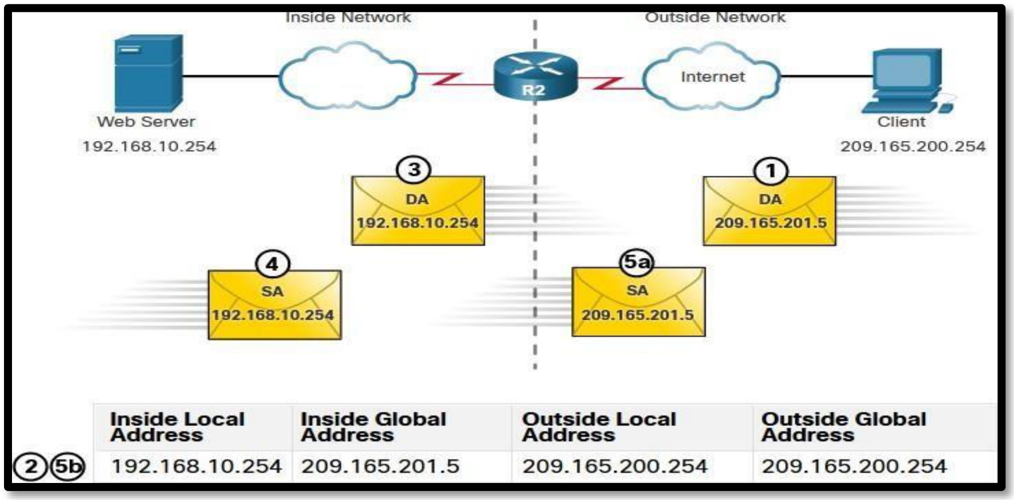
1. Compare Static NAT, Dynamic NAT, and PAT (Port Address Translation).

Static NAT	Dynamic NAT	PAT
one-to-one mapping of local and global addresses	one-to-one mapping of local and global addresses	One Inside Global address (public) can be mapped to many Inside Local addresses (private).
A unique Inside Global address is required for each inside host accessing the outside network.	A unique Inside Global address is required for each inside host accessing the outside network. But it can assign an available public IPv4 address from the pool	A single unique Inside Global address can be shared by many inside hosts accessing the outside network.

2. How does NAT work and why is it used in networking?

- NAT allow a device with a private IPv4 address to access devices and resources outside of the local network, the private address must first be translated to a public address. NAT provides the translation of private addresses to public addresses.
- NAT conserves the legally registered public addressing scheme by allowing the privatization of intranets. It conserves addresses through application port-level multiplexing. NAT provides consistency for internal network addressing schemes. It hides the IPv4 addresses of users and other devices in a local network. It helps with IP address conservation, security, and stable network design.

3. Static NAT allows external devices to initiate connections to internal devices using the statically assigned public address. In the figure, an internal web server may be mapped to a specific inside global address so that it is accessible from outside networks. Analyze how the static NAT translation works between client host 209.165.200.254 and Web Server 192.168.10.254.



(1) The client sends a packet to the web server.

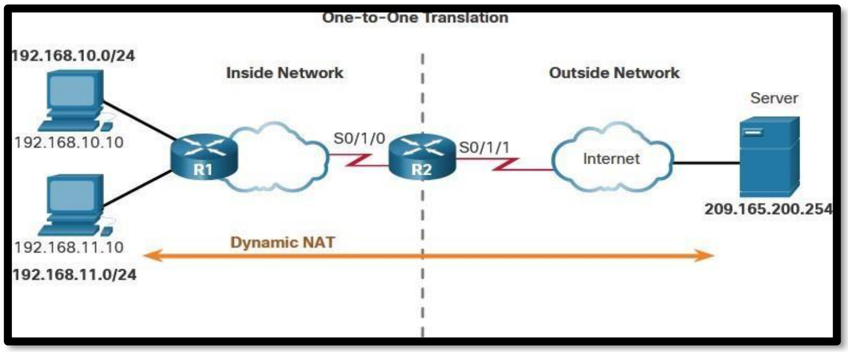
(2) R2 receives packets from the client on its NAT outside interface and checks its NAT table.

(3) R2 translates the inside global address to the inside local address and forwards the packet towards the web server.

(4) The web server receives the packet and responds to the client using its inside local address.

(5) R2 receives the packet from the web server on its NAT inside interface with source address of the inside local address of the web server and translates the source address to the inside global address.

4. In the following scenario, dynamic NAT uses a pool of inside global address 209.165.200.226 and 209.165.200.227. Analyze how Dynamic NAT works between client hosts (192.168.10.10, 192.168.11.10) and Server (209.165.200.254).



(1) PC1 and PC2 send packets requesting a connection to the server.

(2) R2 receives the first packet from PC1, checks the ALC to determine if the packet should be translated, selects an available global address, and creates a translation entry in the NAT table.

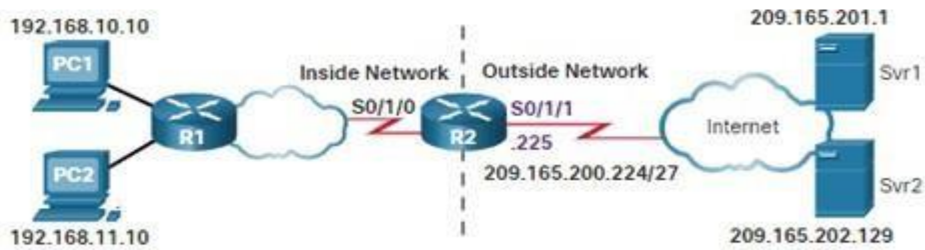
(3) R2 replaces the inside local source address of PC1, 192.168.10.10, with the translated inside global address of 209.165.200.226 and forwards the packet. (The same process occurs for the packet from PC2 using the translated address of 209.165.200.227.)

(4) The server receives the packet from PC1 and responds using the destination address of 209.165.200.226. The server receives the packet from PC2, it responds to using the destination address of 209.165.200.227.

(5) When R2 receives the packet with the destination address of 209.165.200.226; it performs a NAT table lookup and translates the address back to the inside local address 192.168.10.10 and forwards the packet toward PC1.

When R2 receives the packet with the destination address of 209.165.200.227; it performs a NAT table lookup and translates the address back to the inside local address 192.168.11.10 and forwards the packet toward PC2. PC1 and PC2 receive the packet and continue the conversation.

5. In the following scenario, Port address translation (PAT) uses different port numbers 1444 and 1445 to share a single public IP address 209.165.200.225 and distinguish between connections. All hosts from network 192.168.0.0/16 (matching ACL 1) that send traffic through router R2 to the internet. Analyze how PAT works between client hosts (192.168.10.10, 192.168.11.10) and Servers (209.165.201.1, 209.165.202.129).



(1) PC1 and PC2 send packets to Svr1 and Svr2.

(2) The packet from PC1 reaches R2 first. R2 modifies the source IPv4 address to 209.165.200.225 (inside global address). The packet is then forwarded towards Svr1.

(3) The packet from PC2 arrives at R2. PAT changes the source IPv4 address of PC2 to the inside global address 209.165.200.225. PC2 has the same source port number as the translation for PC1. PAT increments the source port number until it is a unique value in its table. In this instance, 1445.

(4) The servers use the source port from the received packet as the destination port, and the source address as the destination address for the return traffic.

(5) R2 changes the destination IPv4 address of the packet from Srv1 from 209.165.200.225 to 192.168.10.10 and forwards the packet toward PC1. R2 changes the destination address of packet from Srv2 from 209.165.200.225 to 192.168.10.11. and modifies the destinations port back to its original value of 1444. The packet is then forwarded toward PC2.